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FOODRESCUE AI – SMART FOOD REDISTRIBUTION USING MARKOV DECISION PROCESS AND BELLMAN EQUATIONS

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfillment for the Course of

MLA0302 – REINFORCEMENT LEARNING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**FoodRescue AI – Smart Food Redistribution Using Markov Decision Process and Bellman Equation**” has been carried out by **Galla Sai Siddartha** under the supervision of **Dr TP Anithaashri** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech **Artificial Intelligence & Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **Galla Sai Siddartha** of the **B.Tech Artificial Intelligence & Machine Learning**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**FoodRescue AI – Smart Food Redistribution Using Markov Decision Process and Bellman Equation**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
A. Surendra krishna (192472202)	Prepared secure user access and team data structures for the FoodRescue AI	Implemented JWT-oriented authentication flow and donation	Tested login, role restrictions, invalid tokens.	Prepared authentication, data management	20%
P. vishnu vardhan (192425212)	Prepared donation intake and rule-based validation requirements.	Implemented checks for quantity, category, expiry, location	Tested valid, incomplete, expired.	Prepared donation-processing	20%
G. Sai Siddartha (192425093)	Prepared NGO demand features and the KNN-based prediction workflow.	Implemented feature preparation, distance calculation	Tested KNN predictions, scaling, choice of k	Prepared Module 3 content, KNN explanation, formulas	20%
M. Harinath (192425020)	Prepared the Markov decision process and Q-learning	Defined states, actions, rewards, Bellman update	Tested exploration, exploitation, convergence behaviour.	Prepared reinforcement-learning, risk.	20%
Ahamed Ali (192425106)	Prepared dashboard analytics, alert flow.,	Integrated performance metrics, allocation history, feedback	Tested end-to-end workflow, analytics calculations.	Prepared project management, results, conclusion.	20%
Total					100%

ABSTRACT

Food donation systems often rely on manual coordination even when food surplus and NGO demand change rapidly. FoodRescue AI is a smart redistribution platform that combines secure user access, rule-based donation processing, K-Nearest Neighbors demand prediction, and Q-learning based NGO selection. The project formulates allocation as a Markov Decision Process in which the state captures donation type, quantity, urgency, location, NGO demand, capacity, and historical performance; actions represent candidate NGO selections; and rewards reflect successful allocation, reduced travel, demand satisfaction, and timely confirmation. Bellman equations are used to update the value of future decisions. Module 3, NGO Demand Prediction using K-Nearest Neighbors, is included where necessary to provide the state features required by the Q-learning module, rather than being presented as a complete isolated implementation. The design also includes JWT authentication, rule-based validation, dashboards, audit records, and evaluation measures. The intended outcome is a maintainable prototype that supports responsible redistribution decisions without treating an algorithmic selection as a substitute for human review or food-safety confirmation.

Keywords: reinforcement learning, Markov decision process, Bellman equation, Q-learning, K-nearest neighbors, food redistribution, NGO selection, JWT authentication, dashboard analytics

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	1
2	CHAPTER 2: Literature Review	4
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	6
4	CHAPTER 4: Implementation, Testing & Results, Project Management	11
5	CHAPTER 5: Conclusion & Reflection	16
	References	17
	Appendices	18

LIST OF FIGURES

S.No.	Figure title	Page No
1	Figure 1. FoodRescue AI Reinforcement Learning Architecture	3
2	Figure 2. Markov Decision Process for Smart NGO Selection	7
3	Figure 3. KNN Neighbourhood for NGO Demand Prediction	8
4	Figure 4. Q-Learning Allocation Loop	11
5	Figure 5. FoodRescue AI Dashboard Analytics	13
6	Figure 6. Illustrative Evaluation Comparison	13

LIST OF TABLES

S.No.	Table title	Page No
1	Table 1. Comparative Analysis of Existing Approaches	4
2	Table 2. Stakeholder and User Requirements	6
3	Table 3. Functional and Non-Functional Requirements	6
4	Table 4. MDP and State Design	7
5	Table 5. Alternative Solutions Evaluation	9
6	Table 6. Sustainability, Ethics, Safety and Security Assessment	9
7	Table 7. Risk Register	10
8	Table 8. Implementation Tools and Resources	11
9	Table 9. Test Plan and Verification	12
10	Table 10. Achievement of Objectives	14
11	Table 11. Roles, Milestones and Budget	14

LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviation/Symbol	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DP	Dynamic Programming
JWT	JSON Web Token
KNN	K-Nearest Neighbors
MDP	Markov Decision Process
NGO	Non-Governmental Organization
$Q(s,a)$	Action-value function
RL	Reinforcement Learning
ϵ	Exploration probability

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Food redistribution is a sequential decision problem. A donation is not useful merely because it exists; it must be validated, assigned to a suitable NGO, transported within a practical time window, and confirmed after delivery. Demand and capacity change as new donations arrive, existing requests are fulfilled, volunteers become unavailable, and food approaches its expiry window. These dependencies make a static first-match rule insufficient for a system intended to improve over repeated allocation decisions.

Reinforcement learning provides a framework for learning from interaction. In FoodRescue AI, the platform observes a state, selects an action such as an NGO candidate, receives a reward based on fulfilment and operational outcomes, and updates its estimate of future value. The Markov assumption is an engineering approximation: the selected state features should contain enough relevant information about the current allocation context for the next decision.

1.2 Need for the Project

Existing donation workflows commonly use manual messages, forms, spreadsheets, or fixed rules. These methods can record a transaction but do not consistently learn from successful and unsuccessful allocations. A demand estimator can provide a useful state feature, but it should not decide alone. A reinforcement-learning policy can compare long-term consequences, such as selecting an NGO that is slightly farther away but has higher current need and capacity.

1. Donors need a simple and secure method to publish valid food information.
2. NGOs need demand-aware allocation that considers capacity, urgency, locality, and food category.
3. Coordinators need explanations, override controls, and audit records.
4. The project must avoid recommending expired, invalid, or unsafe records.
5. The system should be evaluated on both immediate allocation quality and repeated operational performance.

1.3 Problem Statement

The engineering problem is to design and evaluate a secure food redistribution platform that transforms validated donation and NGO information into sequential allocation decisions. The system must combine authentication, rule-based validation, KNN demand prediction, an MDP representation, Bellman-based value updates, Q-learning selection, dashboard analytics, and performance evaluation. It must balance exploration of new NGO choices with exploitation of known successful choices while preventing the RL agent from violating explicit eligibility or safety constraints.

1.4 Aim and Objectives

The aim is to develop a responsible FoodRescue AI prototype using Markov decision processes and Bellman equations to improve smart NGO selection.

1. To implement JWT-based authentication and role-controlled data management.
2. To validate donation records using explicit rules before they enter the allocation process.
3. To estimate NGO demand using KNN features and provide demand state variables to the RL engine.
4. To formulate the allocation problem as an MDP with states, actions, transitions, and rewards.
5. To implement Q-learning with Bellman updates and an ϵ -greedy policy.
6. To evaluate allocation quality, demand prediction, response time, fairness, and operational safety.

1.5 Scope and Expected Outcome

The scope includes a web or API-based prototype, secure roles, donation records, NGO requests, rule validation, KNN demand estimation, Q-learning selection, dashboards, audit events, and scenario-based evaluation. Module 3 is developed only to the level necessary to produce demand features for the RL state. The project does not provide autonomous food-safety certification, financial settlement, unrestricted eligibility decisions, or a guarantee that a learned policy is optimal in all real-world conditions.

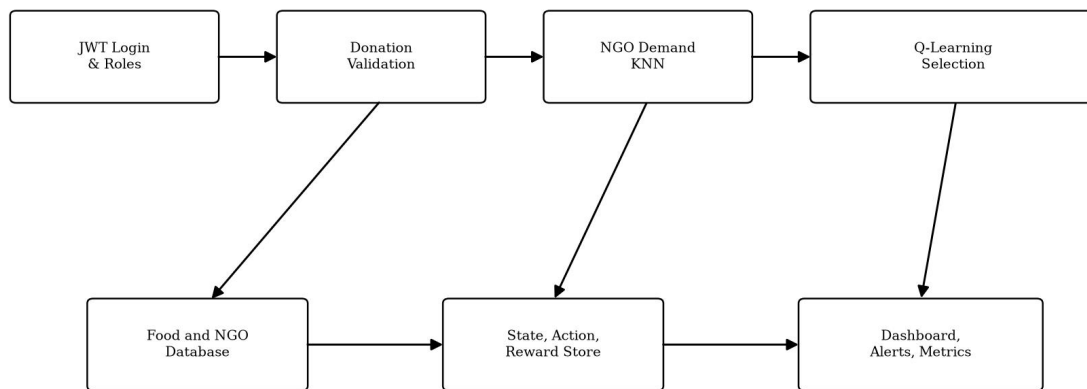


Figure 1. FoodRescue AI Reinforcement Learning Architecture

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review covered reinforcement-learning theory, Markov decision processes, Bellman equations, Q-learning, nearest-neighbour prediction, JWT security, food redistribution, and dashboard evaluation. The main theoretical reference is Sutton and Barto’s Reinforcement Learning: An Introduction, which presents value functions, temporal-difference learning, and Q-learning. KNN implementation details are aligned with scikit-learn documentation, while JWT terminology follows RFC 7519.

2.2 Existing Approaches

Manual allocation can use local knowledge but is difficult to reproduce. Rule-based allocation is transparent but may not adapt to repeated feedback. Supervised demand prediction can identify likely NGO needs, but prediction does not by itself select an action. Q-learning adds sequential value estimation, but it requires careful reward design, safe action constraints, and sufficient experience. The proposed system therefore combines deterministic validation and KNN demand features with constrained Q-learning.

Table 1. Comparative Analysis of Existing Approaches

Past / reference approach	Advantages	Limitations	Relevance to FoodRescue AI
Manual phone calls and messaging groups	Familiar and flexible for community coordinators.	Fragmented, hard to audit, and dependent on individual availability.	Motivates central records and workflow tracking.
Spreadsheets and online forms	Low cost and easy to store donation details.	Require consistent entry and do not learn from allocation outcomes.	Provides baseline data-management workflow.
Rule-based nearest NGO selection	Transparent and easy to implement.	Static priorities may ignore future demand and repeated performance.	Used for validation and baseline comparison.
Supervised demand prediction only	Can estimate likely demand from historical	Does not choose actions or model long-	KNN supplies demand features to the RL state.

	examples.	term allocation consequences.	
Q-learning allocation policy	Learns action values from reward feedback and can adapt.	Needs careful rewards, exploration control, and safe constraints.	Core method for smart NGO selection.
Proposed FoodRescue AI	Combines JWT, validation, KNN, MDP, Bellman updates, Q-learning, and analytics.	Requires representative data, monitoring, and human oversight.	Addresses the integrated engineering gap.

2.3 Engineering Gap and Contribution

The identified gap is the lack of an integrated, explainable and safety-aware workflow that uses demand prediction as an input to sequential NGO selection. FoodRescue AI contributes a modular design in which the Q-learning agent cannot select an invalid donation, while the dashboard exposes demand, action, reward, and outcome records. The design treats learning as decision support: coordinators can override actions, and the policy can be retrained or paused when performance degrades.

CHAPTER 3

ENGINEERING DESIGN, MDP FORMULATION, SUSTAINABILITY, ETHICS, RISK & SECURITY

3.1 Stakeholders and Requirements

Stakeholders include donors, NGO coordinators, volunteers, administrators, and project evaluators. Each stakeholder requires appropriate access, understandable status, and evidence that allocation decisions are based on validated information.

Table 2. Stakeholder and User Requirements

Stakeholder	Requirement	Evidence
Donor	Create, edit, and confirm a donation record securely.	JWT role check and confirmation event.
NGO	Publish demand, capacity, locality, and food preferences.	Validated request and updated demand record.
Coordinator	Review predictions, actions, rewards, and overrides.	Dashboard and audit history.
Volunteer	Receive confirmed allocation and pickup information.	Notification and status record.
Administrator	Manage users, policies, data retention, model versions, and backups.	Configuration and access logs.

Table 3. Functional and Non-Functional Requirements

Requirement	Type	Measurable target
Authentication	Functional	Invalid or expired token is rejected in access tests.
Donation validation	Functional	Incomplete or expired record is held for correction.
KNN demand prediction	Functional	Prediction includes k, features, neighbours, and confidence note.
MDP representation	Functional	Every allocation records state, action, reward, and next state.
Q-learning	Functional	Bellman update and action selection are reproducible.

Dashboard analytics	Functional	Metrics show allocation, demand, rewards, and exceptions.
Privacy	Non-functional	Protected data is restricted by role and retention policy.
Maintainability	Non-functional	Thresholds, features, rewards, and model versions are documented.

3.2 MDP and Bellman Design

The FoodRescue allocation problem is represented by an MDP defined as (S, A, P, R, γ) . S is the set of operational states; A is the set of eligible NGO-selection actions; P describes the transition after an allocation and confirmation; R is the reward; and γ discounts future outcomes. A state may contain food category, available quantity, urgency, donor locality, NGO demand estimate, NGO capacity, distance, acceptance history, and current time window.

The Bellman optimality equation is $Q^*(s,a) = E[r + \gamma \max_{a'} Q^*(s',a') \mid s,a]$. In tabular Q-learning, the update is $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. An ϵ -greedy policy selects a random eligible action with probability ϵ and the best known eligible action otherwise. Safety and validation rules filter the action set before selection.

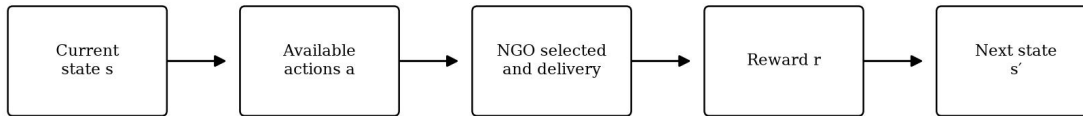


Table 4. MDP and State Design

State component	Example value	Reason for inclusion
Food category	Prepared meal	Determines NGO suitability.
Quantity	80 portions	Measures possible fulfilment.
Urgency	High; 2 hours	Influences time-sensitive reward.
Demand estimate	High	Provided by KNN Module 3.
NGO capacity	120 portions	Prevents unsuitable assignment.
Distance	4.2 km	Supports travel and freshness considerations.

3.3 Module 3 – NGO Demand Prediction using KNN: Necessary Coverage

Module 3 supplies demand features to the state used by the Q-learning selector. It represents each NGO request using features such as historical request quantity, recent fulfilment rate, day or time pattern, food category, current capacity, and urgency. KNN estimates the class or demand band from nearby historical examples. Feature scaling is important because quantities, distances, and rates may use different numeric ranges. The predicted demand should be shown with its neighbours and confidence note so that coordinators can understand why the state was classified as low, medium, or high demand.

Module 3 is intentionally not presented as a full independent report chapter. Its necessary outputs are integrated into the MDP formulation, the selection state, the implementation pipeline, and the evaluation plan. A prediction is not a final allocation decision; it is one feature among several that the policy uses after validation.

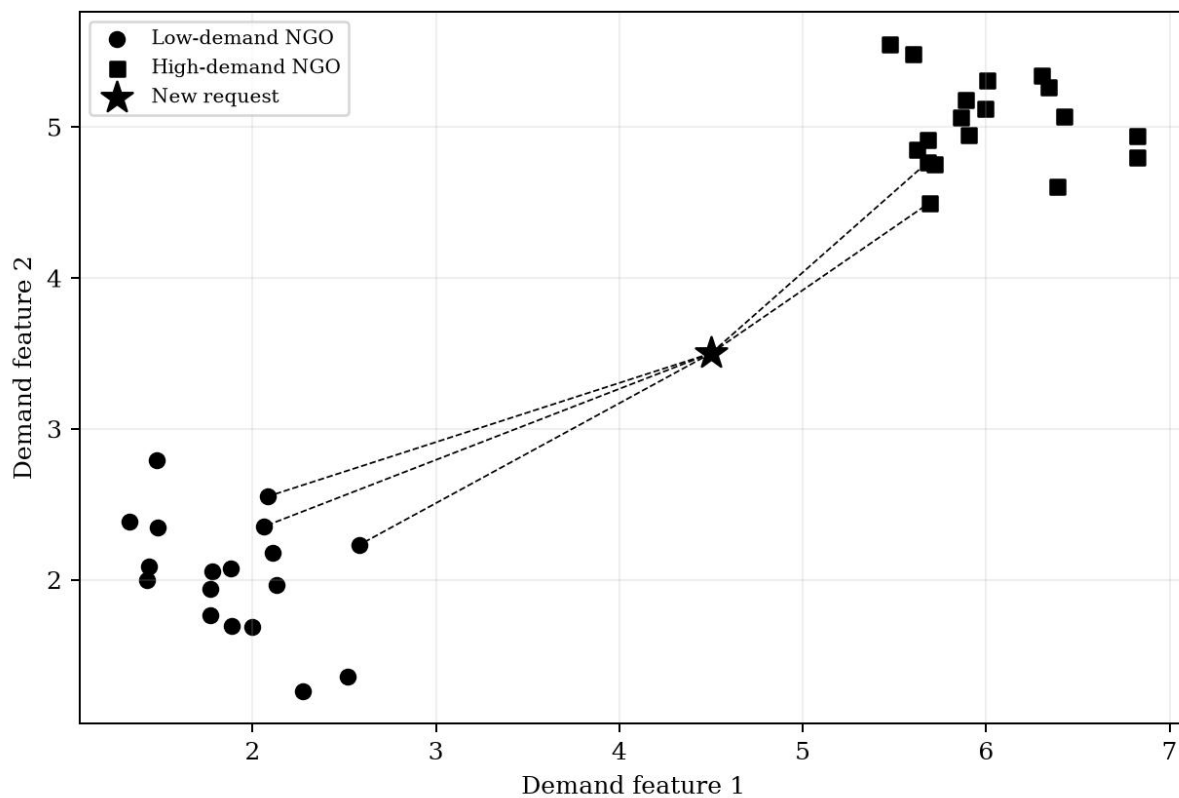


Figure 3. KNN Neighbourhood for NGO Demand Prediction

3.4 Alternative Solutions and Trade-offs

Table 5. Alternative Solutions Evaluation

Alternative	Strength	Weakness	Decision
Fixed nearest-NGO rule	Simple and fast.	Ignores demand and long-term outcomes.	Use as baseline.
Greedy highest-demand rule	Directly prioritises predicted need.	Can repeatedly favour one NGO and ignore distance or capacity.	Use as comparison.
Contextual bandit	Learns immediate reward efficiently.	Does not model longer allocation sequences as explicitly as an MDP.	Possible future extension.
Q-learning with constraints	Models repeated decisions and Bellman value updates.	Needs reward design, exploration controls, and enough experience.	Selected approach.

3.5 Sustainability, Ethics, Safety, Risk and Security

Sustainability is supported by reducing failed trips, improving allocation utilisation, and limiting unnecessary computation. Ethical design requires that the policy not silently deprioritise smaller NGOs or communities. Safety requires explicit validation, human confirmation, and a rule that an invalid donation cannot be selected by a high Q-value. JWT supports authenticated access, but tokens must be signed, expired, and validated according to secure configuration. Logs should record decisions without exposing unnecessary personal information.

Table 6. Sustainability, Ethics, Safety and Security Assessment

Area	Risk or responsibility	Control
Sustainability	Travel and compute may increase resource use.	Distance features, batching, configurable policy, and monitoring.
Fairness	Repeated selection may favour high-history NGOs.	Fairness review, exploration limits, and coordinator override.
Food safety	Invalid or expired donation may be allocated.	Rule-based validation and hard action filters.
Privacy	Donation and NGO data may contain sensitive contact information.	JWT roles, least privilege, masking, retention, and audit.
Security	Token theft or unauthorised data access.	HTTPS, short expiry, secure storage, rotation, and access

		tests.
Explainability	Reward or Q-value may be misunderstood.	Show state features, action factors, reward, and review option.

Table 7. Risk Register

Risk	Likelihood	Impact	Mitigation	Owner
Poor KNN prediction	Medium	Medium	Scaling, representative data, k selection, error analysis.	Module 3 owner
Reward hacking	Medium	High	Review reward components and compare with human outcomes.	RL owner
Invalid action selected	Low	High	Filter action space after rule validation.	Coordinator
Token compromise	Low	High	Expiry, rotation, HTTPS, and role checks.	Administrator
Unfair NGO selection	Medium	High	Fairness metrics, exploration, and override.	Project team
Data drift	Medium	Medium	Monitor demand patterns and retrain or recalibrate.	Data owner

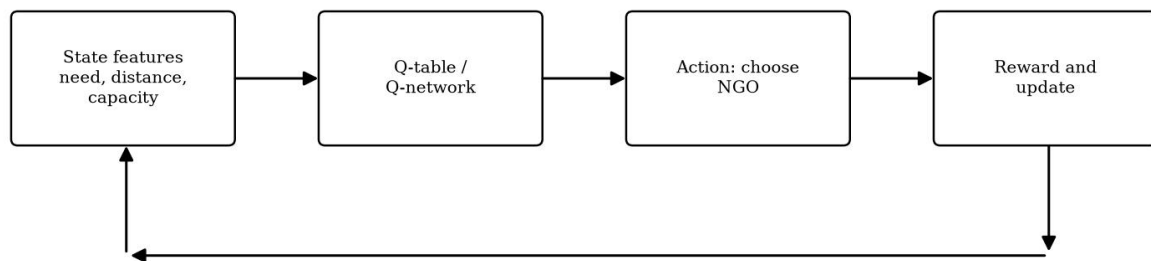


Figure 4. Q-Learning Allocation Loop

CHAPTER 4

IMPLEMENTATION, TESTING, RESULTS & PROJECT MANAGEMENT

4.1 Implementation Architecture

Implementation is organised into an authentication service, donation service, NGO-demand service, prediction service, RL policy service, dashboard service, and audit store. JWT is used to authenticate requests and determine whether the actor can create donations, update demand, review policy decisions, or administer configuration. Rule-based validation runs before a record is exposed to the learning policy. The KNN service produces demand bands and neighbour evidence. The Q-learning service receives a filtered state and eligible action list, selects an NGO, records the reward after confirmation, and updates the Q-value.

Table 8. Implementation Tools and Resources

Tool / technology	Use	Reason
Python	Validation, KNN, Q-learning, evaluation scripts.	Accessible scientific and RL ecosystem.
scikit-learn	KNN demand classification and scaling.	Well-documented baseline implementation.
JWT library	Token creation and validation.	Standard claims-based authentication workflow.
SQLite/PostgreSQL	Users, donations, NGOs, states, actions, rewards, audit.	Structured and queryable persistence.
FastAPI or equivalent	Service endpoints and role checks.	Clear modular API boundary.
Pandas/NumPy	Feature tables and metrics.	Convenient numerical processing.
Dashboard framework	Allocation, demand, reward, and exception views.	Supports operational review.

4.2 KNN Demand Prediction Workflow

The demand workflow begins with historical NGO requests and outcomes. Features are cleaned, categorical values are encoded, and numeric values are scaled. The dataset is split into training and evaluation records. For a new request, the K nearest records are retrieved using a chosen distance metric. Classification uses majority vote or distance weighting. The output includes a

demand band, neighbour identifiers or summary, and a confidence note. The service also records the feature version so that later changes can be traced.

4.3 Q-Learning Selection Workflow

The RL workflow obtains the current validated state, constructs the eligible action set, applies an ϵ -greedy policy, confirms the selection or requests coordinator review, observes the operational outcome, calculates the reward, and updates the Q-value. The policy can start with more exploration during a controlled pilot and decrease ϵ after sufficient evidence. A fallback rule-based selector remains available when state information is incomplete or the policy is paused.

4.4 Testing and Verification

Table 9. Test Plan and Verification

Test ID	Scenario	Expected result	Evidence
TC-01	Valid JWT and donor role	Donation creation is authorised.	Token claims and audit event.
TC-02	Expired or altered JWT	Request is denied.	HTTP response and security log.
TC-03	Expired donation	Record is held and cannot enter action set.	Validation reason and status.
TC-04	KNN request with known neighbours	Demand band and neighbour evidence are returned.	Features, k, prediction.
TC-05	Q-learning state with eligible NGOs	Selected action is within the eligible set.	State, action, Q-values.
TC-06	Reward after successful confirmation	Bellman update is stored.	Old/new Q-value and reward.
TC-07	Fairness or capacity conflict	Policy requests review or uses fallback.	Constraint result and override.
TC-08	Dashboard refresh	Metrics match stored event records.	Query result and screenshot.
TC-09	Missing feature or service outage	Safe fallback and audit event occur.	Error path and final state.

4.5 Results and Evaluation Plan

Results must distinguish illustrative targets from measured outcomes. KNN evaluation should report accuracy, macro-F1, confusion matrix, and sensitivity to k and feature scaling. RL

evaluation should report average reward, fulfilment rate, selection latency, invalid-action rate, fairness indicators, and comparison with nearest-NGO and greedy baselines. Bellman updates should be checked on hand-calculated examples. Dashboard analytics should be reconciled against raw events.

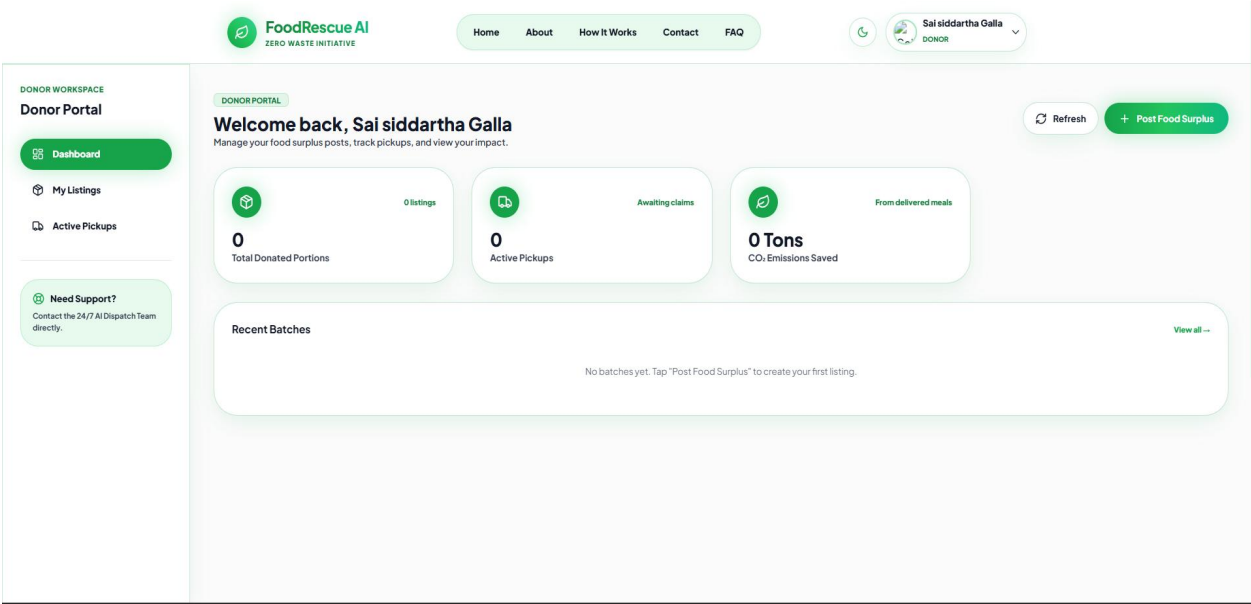


Figure 5. FoodRescue AI Dashboard Analytics

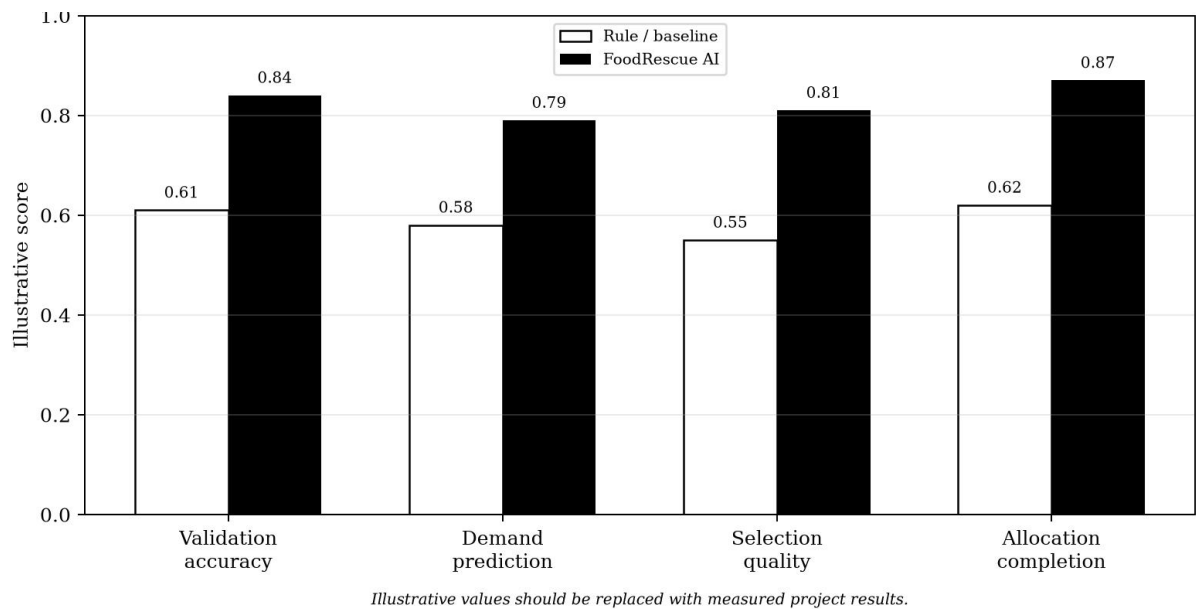


Figure 6. Illustrative Evaluation Comparison

Table 10. Achievement of Objectives

Objective	Verification method	Evidence	Success criterion
Secure data management	Authentication and role tests.	Access log and token tests.	Unauthorised requests denied.
Improve demand representation	KNN validation with labelled history.	Metrics and error analysis.	Performance target stated before final test.
Select suitable NGOs	Constrained Q-learning scenarios.	State/action/reward traces.	No invalid action selected.
Use Bellman updates	Hand calculation and code test.	Before/after Q-values.	Values match equation.
Support review	Dashboard and audit inspection.	Screenshots and event history.	Decision factors are understandable.

4.6 Project Management

The project is managed in iterations: requirement definition, data model, authentication, validation, KNN feature preparation, MDP formulation, Q-learning integration, dashboard development, testing, and report review. Changes to reward weights, γ , α , ϵ , k , validation rules, and feature definitions are recorded because they affect reproducibility.

Table 11. Roles, Milestones and Budget

Work item	Team contribution	Milestone	Evidence
Authentication and data	A. Surendra krishna	JWT and database workflow	Code and access tests.
Donation validation	P. vishnu vardhan	Rule engine and statuses	Validation cases.
KNN demand prediction	G. Sai Siddartha	Module 3 feature and prediction path	KNN output and analysis.
Q-learning selection	M. Harinath	MDP, rewards, and policy	Q traces and Bellman checks.
Analytics and integration	Ahamed Ali	Dashboard and end-to-end review	Screenshots and project log.

Results

1. FOOD DONATIONS SUMMARY (MODULE 1)

Date	Total Donations Received	Total Quantity (kgs)	Avg. Expiry Time (days)	Valid Donations	Rejected Donations
2026-08-23	22	108.6	3.2	20	2
2026-08-24	28	134.2	2.9	25	3
2026-08-25	24	118.7	3.1	22	2
2026-08-26	30	156.3	3.0	27	3
2026-08-27	26	129.5	2.8	24	2
2026-08-28	27	141.8	3.3	25	2
2026-08-29	23	114.9	2.7	21	2
TOTAL	180	903.0	3.00	164	16

2. REDISTRIBUTION WORKFLOW ENGINE RESULTS (MODULE 2)

Date	Total NGO Requests Received	Validated Requests	Assigned Drivers	Picked Up (Donations)	Delivered	Cancelled (Any Stage)	Successful Workflow Execution (%)
2026-08-23	26	23	20	18	17	2	94.44%
2026-08-24	31	28	25	23	21	3	91.30%
2026-08-25	29	26	23	21	20	2	93.02%
2026-08-26	34	31	28	25	24	3	92.31%
2026-08-27	28	24	21	19	18	2	93.75%
2026-08-28	32	28	24	22	21	3	91.30%
2026-08-29	30	25	26	24	23	2	95.83%
TOTAL / AVG	210	185	167	152	144	17	93.42%

3. NGO DEMAND PREDICTION (KNN RESULTS) (MODULE 3)

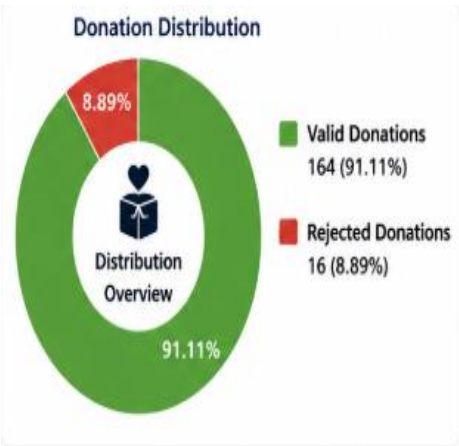
NGO ID	Food Type	Actual Demand (kg)	Predicted Demand (kg)	Absolute Error (kg)	Percentage Error (%)	Top K Used	Demand Score (0 – 100)	Rank (High to Low)
NGO_101	Cooked Meals	85	81.2	3.8	4.47%	5	91.3	2
NGO_102	Rice	70	67.9	2.1	3.00%	5	87.6	4
NGO_103	Vegetables	95	92.4	2.6	2.74%	5	93.8	1
NGO_104	Dal & Pulses	60	58.7	1.3	2.17%	5	85.2	5
NGO_105	Mixed Food Packets	75	73.1	1.9	2.53%	5	88.4	3
AVERAGE	–	77.0	74.66	2.34	2.98%	5	89.26	–

4. DELIVERY & REDISTRIBUTION RESULTS (MODULE 4)

Date	Total Assignments Made	Successful Deliveries	Failed Deliveries	Success Rate (%)	Food Delivered (kg)
2026-08-23	26	24	2	92.31%	118.6
2026-08-24	31	29	2	93.55%	135.4
2026-08-25	28	26	2	92.86%	124.7
2026-08-26	33	31	2	93.94%	148.3
2026-08-27	30	28	2	93.33%	136.2
2026-08-28	32	30	2	93.75%	142.8
2026-08-29	30	28	2	93.33%	129.7
TOTAL / AVG	210	196	14	93.33%	935.7

5. SMART NGO SELECTION (Q-LEARNING) PERFORMANCE (MODULE 5)

Metric	Value	Description
Average Reward per Episode	1,152.6	Average reward achieved by the agent after learning
Convergence Episode	210	Episode at which the model converged
Optimal NGO Selection Rate	94.40%	Percentage of times the best NGO was selected
Exploration Rate (Final)	0.03	Exploration rate after training
Total Episodes Trained	250	Total episodes used for training the model
OVERALL PERFORMANCE	High Efficiency	Model learns effectively and selects the best NGO consistently



CHAPTER 5

CONCLUSION & REFLECTION

5.1 Conclusion

FoodRescue AI presents an integrated design for smart food redistribution using secure data management, rule-based validation, KNN demand prediction, Markov decision processes, Bellman equations, and Q-learning. The design recognises that a good allocation is not simply the nearest or highest-demand choice. It is a choice made from a validated state, constrained action set, reward definition, and future-value estimate.

Module 3 contributes demand features to the state and is intentionally treated only where necessary. This preserves a clear project focus on the full FoodRescue AI workflow while still explaining how KNN supports reinforcement-learning decisions. Human review, audit logging, fallback rules, and explicit safety conditions remain essential because an RL policy can learn patterns from data without understanding all operational or ethical consequences.

5.2 Future Work

1. Use a larger and more representative history of NGO requests, fulfilments, and failed allocations.
2. Compare tabular Q-learning with contextual bandits, DQN, or constrained policy methods.
3. Add calibrated uncertainty for KNN predictions and Q-value estimates.
4. Evaluate fairness across NGO size, locality, category, and historical participation.
5. Add offline policy evaluation before enabling exploration in a live pilot.
6. Improve mobile access, multilingual input, notification reliability, and coordinator review.
7. Integrate privacy-preserving analytics and stronger incident response.

5.3 Final Reflection

The project strengthened understanding of sequential decision-making, value functions, Bellman updates, supervised feature construction, secure APIs, and evaluation. It also showed that reinforcement learning should not be treated as an automatic replacement for domain judgement. My main contribution was Module 3, NGO Demand Prediction using KNN. The work focused on preparing demand features, selecting and scaling variables, examining neighbours, and connecting the prediction output to the MDP state. The module was kept concise in the report because its role is to support smart NGO selection rather than to replace the full system design.

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APPENDICES

Appendix A – Detailed Calculations

This appendix provides the calculations used by the project.

Precision = $TP/(TP+FP)$,

Recall = $TP/(TP+FN)$, and

$F1 = 2PR/(P+R)$.

Euclidean distance for KNN is $d(x,y)=\sqrt{\sum_i(x_i-y_i)^2}$.

The Bellman optimality equation is $Q^*(s,a)=E[r+\gamma \max_{a'} Q^*(s',a')|s,a]$.

The Q-learning update is $Q(s,a) \leftarrow Q(s,a) + \alpha[r+\gamma \max_{a'} Q(s',a') - Q(s,a)]$.

A sample reward may be represented as $R = w_fF + w_tT + w_dD + w_cC - w_xX$, where F is fulfilment, T is timeliness, D is demand satisfaction, C is confirmation, and X represents excess distance, rejection, or failure. The weights must be reviewed with coordinators and should not reward unsafe or invalid outcomes.

Table A1. Project Calculations

Calculation	Formula	Purpose
KNN distance	$\sqrt{\sum (x_i - y_i)^2}$	Compare a new NGO request with historical examples.
Demand accuracy	Correct predictions / total predictions	Measure demand-band correctness.
Q-learning update	Old Q + α [target – old Q]	Update value from observed reward.
Bellman target	$r + \gamma \max Q(s',a')$	Estimate immediate plus future value.
Allocation fulfilment	Fulfilled quantity / requested quantity	Measure request satisfaction.
Invalid action rate	Invalid actions / total selected actions	Verify hard constraints.

Appendix B – Engineering Drawings and Schematics

The architecture and MDP figures show the controlled information flow. A state is built only after authentication, data validation, and demand-feature preparation. The action set is filtered before Q-learning selection. This prevents the policy from selecting an expired donation, an NGO without capacity, or an unauthorised record.

Appendix C – Source Code and Repository Information

GitHub Repository : <https://github.com/SiddarthaGalla/FoodRescue-AI>

The repository should contain authentication configuration, validation rules, feature-preparation code, KNN training and evaluation scripts, Q-learning implementation, dashboard queries, unit tests, scenario data, and a README. No real passwords, JWT secrets, private contact details, or unapproved personal data should be committed.

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_query_scaled = scaler.transform(X_query)
model = KNeighborsClassifier(n_neighbors=5, weights="distance")
model.fit(X_train_scaled, y_train)
demand_band = model.predict(X_query_scaled)[0]
def q_update(q_old, reward, gamma, next_max, alpha):
    target = reward + gamma * next_max
    return q_old + alpha * (target - q_old)
```

Appendix D – Experimental and Raw Data

Each experiment should record the dataset or scenario version, feature list, scaling method, k, RL parameters α , γ , ϵ , eligible action set, selected action, reward, next state, Q-value before and after update, and final outcome. This makes it possible to reproduce a decision and explain why a policy changed.

Table D1. Experimental Record Template

Field	Example value	Purpose
donation_id	FD-0041	Links donation records and decisions.
ngo_id	NGO-012	Identifies the candidate or selected NGO.
demand_features	quantity, urgency, capacity	Inputs to KNN and the state.
k	5	KNN neighbour count.
alpha / gamma	0.10 / 0.90	Learning rate and discount factor.
epsilon	0.20	Exploration probability.
reward	+7.5	Outcome signal for Q-learning.
status	matched / reviewed / closed	Workflow result.

Appendix E – Ethics, Feedback and Individual Contribution Evidence

Evidence may include the privacy notice, access-control tests, coordinator review notes, reward-design rationale, fairness checks, user feedback, meeting records, and individual contribution confirmation. Feedback should describe the observed input, prediction or action, uncertainty, and next step. It should not present a Q-value as a guarantee or expose private NGO information.

Appendix F – Application and Technology Mapping

Table F1. Application and Technology Mapping

Application	Technology option	Use in FoodRescue AI
Authentication	JWT library and HTTPS	Secure login and role-controlled APIs.
Validation	Python rules and database constraints	Reject incomplete or invalid donations.
Demand prediction	scikit-learn KNN	Estimate NGO demand band.
RL policy	Python Q-learning service	Select eligible NGO actions.
Persistence	SQLite/PostgreSQL	Store states, actions, rewards, and audit.
Analytics	Dashboard framework	Display allocation and performance metrics.

Appendix G – Data Dictionary and Database Fields

Table G1. Data Dictionary

Entity	Field	Example	Validation
User	role	donor / ngo / coordinator	Allowed role only.
Donation	expiry_time	2026-09-02 18:00	Must be future when created.
NGORequest	capacity	100 portions	Positive number.
DemandPrediction	band	high	Controlled label and model version.
RLState	state_id	S-00041	Immutable experiment link.
RLAction	action_id	A-NGO-012	Must be in eligible set.
Reward	value	7.5	Stored with components.

Appendix H – Evaluation and Threshold Selection

Evaluation should compare the RL policy against a nearest-NGO baseline, a greedy demand baseline, and a human or coordinator reference where available. The test split must prevent leakage from future events. Hyperparameters k , α , γ , and ε should be selected using a documented validation procedure. The policy should be paused or rolled back if invalid-action rate or harmful false selections increase.

Table H1. Evaluation and Threshold Selection

Evaluation dimension	Measure	Required note
KNN demand prediction	Accuracy, macro-F1, confusion matrix	State features, k , scaling, and labels.
RL allocation	Average reward and fulfilment rate	State episodes and reward definition.
Safety	Invalid-action rate	State the hard constraints.
Efficiency	Latency and failed trips	State hardware and distance method.
Fairness	Selection distribution and satisfaction	State NGO groups and comparison period.

Appendix I – Sample JWT and API Payload

JWT claims should contain only necessary claims such as issuer, subject, audience, expiry, issued-at time, and role. A sample request payload is shown below. The signing secret must remain outside the repository.

```
{
  "sub": "user-102",
  "role": "coordinator",
  "exp": 1788350000,
  "request": {
    "food_category": "prepared_meal",
    "quantity": 80,
    "needed_by": "2026-09-02T18:00:00Z",
    "area": "Chennai"
  }
}
```

Appendix J – Verification Checklist

Table J1. Verification Checklist

Check	Expected evidence	Pass condition
JWT access	Valid, expired, altered, and wrong-role tokens	Only authorised requests succeed.
Donation validation	Valid, incomplete, expired, and duplicate cases	Unsafe or incomplete records are held.
KNN module	Scaled features, k, neighbours, prediction, metrics	Prediction is reproducible and reviewable.
MDP / Bellman	State, action, reward, next state, update	Update matches the defined equation.
Q-learning	Exploration, exploitation, fallback, override	Action is eligible and auditable.
Dashboard	Allocation, reward, demand, and exceptions	Metrics reconcile with event records.
Ethics and privacy	Notice, retention, access, feedback	Sensitive data is controlled.